

Database Design Document

AssetFlow AI

Database Architecture

Version: 1.0

Database: PostgreSQL

ORM: Prisma

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1. Database Overview

AssetFlow AI uses a relational PostgreSQL database.

Goals

- Fast Search
- High Integrity
- ACID Transactions
- Easy Reporting
- AI-ready Data

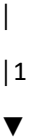
2. Entity Relationship Diagram

```text

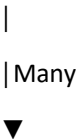
Departments



Users

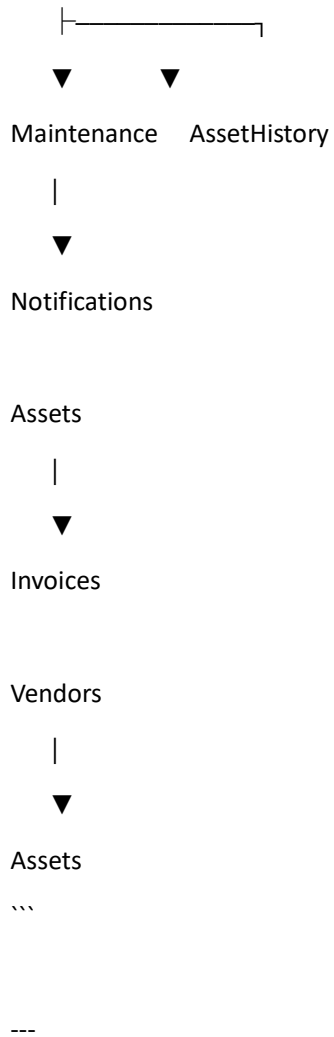


Assignments



Assets





### # 3. Naming Convention

#### Table Names

snake\_case

#### Example

users

assets

maintenance

Primary Key

id

Foreign Keys

user\_id

asset\_id

department\_id

Timestamps

created\_at

updated\_at

deleted\_at

---

# 4. Tables

---

## users

Purpose

Stores all system users.

Columns

| Column        | Type         |
|---------------|--------------|
| id            | UUID         |
| full_name     | VARCHAR(100) |
| email         | VARCHAR(255) |
| password_hash | TEXT         |
| role          | ENUM         |
| phone         | VARCHAR(20)  |
| department_id | UUID         |
| profile_image | TEXT         |
| status        | BOOLEAN      |
| created_at    | TIMESTAMP    |
| updated_at    | TIMESTAMP    |
| deleted_at    | TIMESTAMP    |

---

## departments

| Column      | Type      |
|-------------|-----------|
| id          | UUID      |
| name        | VARCHAR   |
| description | TEXT      |
| manager_id  | UUID      |
| location    | VARCHAR   |
| created_at  | TIMESTAMP |
| updated_at  | TIMESTAMP |

---

## ## vendors

| Column      | Type      |
|-------------|-----------|
| id          | UUID      |
| vendor_name | VARCHAR   |
| email       | VARCHAR   |
| phone       | VARCHAR   |
| address     | TEXT      |
| gst_number  | VARCHAR   |
| created_at  | TIMESTAMP |

---

## ## assets

### Purpose

Stores every asset.

| Column        | Type    |
|---------------|---------|
| id            | UUID    |
| asset_code    | VARCHAR |
| asset_name    | VARCHAR |
| category      | VARCHAR |
| serial_number | VARCHAR |
| model         | VARCHAR |
| manufacturer  | VARCHAR |

|                          |
|--------------------------|
| purchase_date   DATE     |
| purchase_price   DECIMAL |
| warranty_expiry   DATE   |
| condition   ENUM         |
| status   ENUM            |
| location   VARCHAR       |
| department_id   UUID     |
| vendor_id   UUID         |
| qr_code   TEXT           |
| image_url   TEXT         |
| created_at   TIMESTAMP   |
| updated_at   TIMESTAMP   |
| deleted_at   TIMESTAMP   |

---

## ## assignments

Tracks which employee currently owns an asset.

|                        |
|------------------------|
| Column   Type          |
| ----- -----            |
| id   UUID              |
| asset_id   UUID        |
| user_id   UUID         |
| assigned_by   UUID     |
| assigned_date   DATE   |
| expected_return   DATE |
| returned_date   DATE   |
| remarks   TEXT         |

---

## ## maintenance

| Column        | Type      |
|---------------|-----------|
| id            | UUID      |
| asset_id      | UUID      |
| technician_id | UUID      |
| issue         | TEXT      |
| priority      | ENUM      |
| status        | ENUM      |
| repair_cost   | DECIMAL   |
| opened_at     | TIMESTAMP |
| completed_at  | TIMESTAMP |

---

## ## asset\_history

Stores complete lifecycle.

| Column      | Type      |
|-------------|-----------|
| id          | UUID      |
| asset_id    | UUID      |
| action      | VARCHAR   |
| user_id     | UUID      |
| description | TEXT      |
| created_at  | TIMESTAMP |



Example Actions

- Created
- Assigned
- Returned
- Maintenance Started
- Maintenance Completed
- Disposed

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## invoices

| Column         | Type    |
|----------------|---------|
| id             | UUID    |
| asset_id       | UUID    |
| invoice_number | VARCHAR |
| invoice_date   | DATE    |
| amount         | DECIMAL |
| pdf_url        | TEXT    |

---

## notifications

| Column  | Type    |
|---------|---------|
| id      | UUID    |
| user_id | UUID    |
| title   | VARCHAR |

|  |            |  |           |  |
|--|------------|--|-----------|--|
|  | message    |  | TEXT      |  |
|  | is_read    |  | BOOLEAN   |  |
|  | created_at |  | TIMESTAMP |  |

---

## ## audit\_logs

Tracks every sensitive activity.

|  |            |  |           |  |
|--|------------|--|-----------|--|
|  | Column     |  | Type      |  |
|  | -----      |  | -----     |  |
|  | id         |  | UUID      |  |
|  | user_id    |  | UUID      |  |
|  | action     |  | VARCHAR   |  |
|  | table_name |  | VARCHAR   |  |
|  | record_id  |  | UUID      |  |
|  | old_value  |  | JSONB     |  |
|  | new_value  |  | JSONB     |  |
|  | ip_address |  | VARCHAR   |  |
|  | created_at |  | TIMESTAMP |  |

---

## # 5. Relationships

users

↓

belongs to



departments



assets



belongs to



vendors



assets



belongs to



departments



assignments



belongs to



users



assignments



belongs to



assets



maintenance



belongs to



assets

---

maintenance

↓

belongs to

↓

technicians(users)

---

asset\_history

↓

belongs to

↓

assets

---

notifications

↓

belongs to

↓

users

---

# 6. Enums

Role

```
```text
```

ADMIN

MANAGER

TECHNICIAN

EMPLOYEE

```
```
```

Asset Status

```
```text
```

AVAILABLE

ASSIGNED

MAINTENANCE

LOST

DISPOSED

```
```
```

Condition

```text

NEW

GOOD

FAIR

DAMAGED

SCRAP

```

Priority

```text

LOW

MEDIUM

HIGH

CRITICAL

```

Maintenance Status

```text

OPEN

IN_PROGRESS

COMPLETED

CANCELLED

```

---

# 7. Indexing Strategy

Indexes improve search performance.

Create indexes on

email

asset\_code

serial\_number

department\_id

vendor\_id

status

purchase\_date

warranty\_expiry

Example

```
```sql
CREATE INDEX idx_assets_status
ON assets(status);
```
```

---

# 8. Constraints



Unique

email

asset\_code

serial\_number

invoice\_number

Foreign Keys

department\_id

vendor\_id

asset\_id

user\_id

Check Constraints

purchase\_price > 0

repair\_cost >= 0

expected\_return >= assigned\_date

---

## # 9. Audit Fields

Every important table contains

created\_at

updated\_at

deleted\_at

This enables

- Tracking
- Recovery
- Compliance

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## # 10. Soft Delete

Instead of deleting records

Set

deleted\_at = CURRENT\_TIMESTAMP

Advantages

- Recover data
- Preserve history
- Better audit support

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## # 11. Sample Asset Record

```
``json
{
 "asset_code": "AST-000124",
 "asset_name": "Dell Latitude 7440",
 "category": "Laptop",
 "serial_number": "DL7440X992",
 "manufacturer": "Dell",
 "purchase_price": 85000,
 "status": "ASSIGNED",
 "condition": "GOOD",
 "department_id": "uuid",
 "vendor_id": "uuid"
}
``
```

---

## # 12. Suggested Prisma Model (Example)

```
``prisma
model Asset {
 id String @id @default(uuid())
 assetCode String @unique
 assetName String
 category String
 serialNumber String @unique
}
```

manufacturer String  
purchasePrice Decimal  
purchaseDate DateTime  
warrantyExpiry DateTime  
status AssetStatus  
condition AssetCondition  
qrCode String?  
imageUrl String?

departmentId String  
vendorId String

department Department @relation(fields: [departmentId], references: [id])  
vendor Vendor @relation(fields: [vendorId], references: [id])

createdAt DateTime @default(now())  
updatedAt DateTime @updatedAt  
deletedAt DateTime?

}

...

---

### # 13. Future Expansion

Additional tables planned

- asset\_categories
- purchase\_orders
- depreciation
- insurance

- warranties
- RFID\_logs
- GPS\_tracking
- IoT\_sensor\_data
- AI\_predictions
- chatbot\_history
- approval\_workflows

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## # Database Design Principles

✓ Fully Normalized (3NF)

✓ UUID Primary Keys

✓ Foreign Key Constraints

✓ Indexed Search Fields

✓ Soft Deletes

✓ Audit Logging

✓ AI-Compatible Structure

✓ Scalable for 100,000+ Assets

✓ Production Ready

✓ Prisma ORM Friendly